

Data Cleaning and Preprocessing Report

Project Title

Data Cleaning and Preprocessing – Customer Personality Analysis Dataset

1. Introduction

Data preprocessing is one of the most important stages in any Machine Learning project. A dataset collected from different sources often contains missing values, duplicate records, incorrect data types, inconsistent categories, invalid dates, and outliers. If these problems are not handled properly, they can reduce the accuracy of machine learning models and produce misleading business insights.

The objective of this project was to clean, preprocess, validate, and prepare the Customer Personality Analysis dataset for future Exploratory Data Analysis (EDA) and Machine Learning tasks. All preprocessing steps were implemented using Python, making the workflow fully reproducible.

2. Dataset Overview

Dataset Name: Customer Personality Analysis

File Name: marketing_campaign.csv

Programming Language: Python 3.x

Development Environment: Python IDLE

Libraries Used:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

The dataset contains customer demographic information, purchasing behavior, campaign responses, and income details. These attributes help businesses understand customer personalities and improve marketing strategies.

3. Objectives

The main objectives of this project were:

- Load and inspect the dataset.
- Identify data quality issues.
- Handle missing values.
- Remove duplicate records.
- Correct incorrect data types.
- Convert date columns into datetime format.
- Standardize categorical variables.
- Validate customer ages.
- Detect and handle outliers.
- Validate the cleaned dataset.
- Export the cleaned dataset for future analysis.

4. Dataset Inspection

The dataset was initially explored to understand its overall structure.

The following checks were performed:

- Displayed the first 10 rows.
- Displayed the last 10 rows.
- Examined the dataset shape.
- Reviewed column names.
- Checked data types.
- Generated summary statistics using descriptive analysis.

This initial inspection helped identify potential data quality problems before preprocessing.

5. Data Quality Assessment

A complete data quality assessment was performed to identify issues that could affect data analysis and machine learning.

The following problems were investigated:

- Missing values
- Duplicate rows
- Duplicate customer IDs
- Incorrect data types
- Invalid date values
- Inconsistent categorical values
- Unrealistic customer ages
- Numerical outliers

Each issue was analyzed before applying an appropriate preprocessing technique.

6. Handling Missing Values

The dataset contained missing values, primarily in the **Income** column.

The following steps were performed:

- Counted missing values in every column.
- Calculated missing value percentages.
- Analyzed the distribution of Income.
- Filled missing Income values using the **median**.

Justification

Median imputation was selected because income data usually contains extreme values (outliers). Unlike the mean, the median is less affected by unusually high or low incomes and therefore provides a more reliable estimate.

7. Duplicate Removal

The dataset was checked for:

- Duplicate rows
- Duplicate customer IDs

Duplicate records were removed while preserving the first valid occurrence.

This ensured that every customer was represented only once in the dataset.

8. Data Type Correction

Several columns were reviewed to ensure correct data types.

Examples include:

Variable	Correct Data Type
ID	Integer
Income	Float
Recency	Integer
Dt_Customer	DateTime

Correct data types improve processing speed and ensure accurate calculations.

9. Date Conversion

The **Dt_Customer** column was converted into the DateTime format.

Additional features extracted from the enrollment date included:

- Join Year
- Join Month
- Join Day

Invalid dates were identified and handled during conversion.

These new features can be useful for future customer behavior analysis.

10. Standardization of Categorical Variables

Categorical columns such as **Education** and **Marital_Status** were standardized.

The following preprocessing steps were applied:

- Removed unnecessary spaces.
- Converted text into a consistent format.
- Corrected inconsistent spellings.
- Merged similar categories where appropriate.

Examples:

- "single", "Single", and " SINGLE " were standardized to **Single**.
- Similar marital status values were merged to improve consistency.

This step improves data quality and prevents duplicate categories during analysis.

11. Customer Age Validation

Customer age was calculated using:

Age = Current Year – Year_Birth

The following unrealistic records were identified:

- Negative ages
- Future birth years
- Customers older than 100 years
- Customers younger than 18 years

Such records were considered unrealistic and removed to improve dataset reliability.

12. Outlier Detection

Several numerical variables were analyzed, including:

- Income
- MntWines
- MntFruits
- MntMeatProducts
- MntFishProducts
- MntSweetProducts
- MntGoldProds

The following techniques were used:

- Boxplots
- Interquartile Range (IQR) Method

Outlier summary reports and boxplot visualizations were generated for each variable.

13. Outlier Handling

After detecting outliers, each variable was carefully analyzed.

Instead of removing valuable customer information, **Winsorization (IQR Capping)** was applied.

Reason

High-income customers and premium buyers may represent real business cases rather than incorrect data. Therefore, extreme values were capped within acceptable limits instead of deleting those records.

This approach preserves valuable information while reducing the influence of extreme observations.

14. Dataset Validation

After completing all preprocessing steps, the dataset was validated.

The following checks were performed:

- Missing values verified.
- Duplicate rows removed.

- Duplicate customer IDs removed.
- Data types verified.
- Dates validated.
- Categories standardized.
- Outliers handled.

The final dataset passed all validation checks and was considered ready for further analysis.

15. Before and After Comparison

Metric	Before Cleaning	After Cleaning
Dataset Status	Raw	Clean
Missing Values	Present	Handled
Duplicate Records	Present	Removed
Invalid Data Types	Present	Corrected
Invalid Dates	Present	Corrected
Inconsistent Categories	Present	Standardized
Unrealistic Ages	Present	Removed
Outliers	Present	Handled
Dataset Quality	Low	High

16. Challenges Faced

Several challenges were encountered during preprocessing:

- Missing values in the Income column.
- Identifying the best method for missing value imputation.
- Handling inconsistent categorical values.
- Detecting unrealistic customer ages.
- Managing numerical outliers without losing important business information.
- Ensuring that all preprocessing steps remained reproducible.

These challenges were resolved using appropriate preprocessing techniques and validation procedures.

17. Final Dataset Quality

After preprocessing, the dataset became:

- Clean
- Consistent
- Reliable

- Well-structured
- Machine Learning Ready
- Suitable for Exploratory Data Analysis

The cleaned dataset is expected to improve model performance and produce more accurate business insights.

18. Recommendations

For future work, the following improvements are recommended:

- Perform feature engineering.
- Normalize numerical variables if required.
- Apply feature selection techniques.
- Conduct detailed Exploratory Data Analysis (EDA).
- Build classification and clustering models.
- Evaluate different machine learning algorithms for customer segmentation and campaign prediction.

19. Conclusion

The Customer Personality Analysis dataset was successfully cleaned and preprocessed using Python. All major data quality issues, including missing values, duplicate records, incorrect data types, inconsistent categories, unrealistic ages, and outliers, were addressed systematically.

The preprocessing workflow is fully reproducible and prepares the dataset for future exploratory analysis and machine learning applications. The final cleaned dataset is accurate, consistent, and suitable for predictive modeling and business decision-making.